Prenatal or Postnatal?

The long-run damage of early-life violence, from the end of paramilitary violence in Colombia

Idea 1: statement and assessment

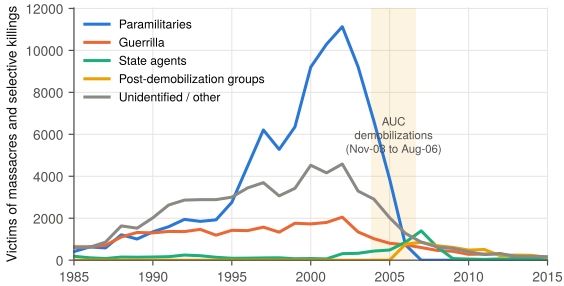
October 2026

Question

Is the long-run damage of early-life exposure to violence prenatal or postnatal?

- If prenatal: the channel is maternal stress in pregnancy (fetal origins). Policy targets pregnant women.
- If postnatal: the channel is the child's environment after birth (caregiver, household, direct exposure). Policy targets early childhood.
- Today only the prenatal effect is credible: León (2012, *JHR*) found both, and its full-census replication (Huaroto, Gonzales and Pacheco, 2026, *JHR*) keeps only the in-utero effect.

Setting: a violence that ends on a known date



- Paramilitary blocs demobilize one by one, November 2003 to August 2006; zero victims after 2007.
- For each municipality m we observe the month T_m when violence ended.
- Children born in the same municipality differ in their age at T_m: born after it, never exposed; born just before, exposed only in utero; born earlier, exposed also in infancy or preschool.

Best specification

Person i born in municipality m in cohort c; $k = c - T_m$ is age at the end of violence.

$$Y_{imc} = \sum_{k \neq -1} \beta_k \mathbf{1}[c - T_m = k] + \mu_m + \lambda_{c \times d(m)} + \varepsilon_{imc}$$

- Reference: cohorts born the year after the end (never exposed). β_0: exposed only in utero. β_1, β_2: also in infancy. β_3 to β_5: also in preschool.
- **Test:** $\beta_0 \neq 0$ and $\beta_1 = \beta_2 = \beta_0$ means prenatal only; β_k rising with k means postnatal damage.
- Estimated with Callaway and Sant'Anna (2021) or de Chaisemartin and D'Haultfœuille (2024) for staggered timing; pre-conception placebos; Rambachan and Roth (2023) bounds.
- Location fixed before the shock: municipality of birth. Outcomes at 20–36: schooling, employment, earnings (GEIH), or test scores at 17 (Saber 11).

Contribution

- The first separation of prenatal from postnatal damage of violence with an exogenous end date, set by a national negotiation rather than by local conditions.
- A mechanism test: by displacement of the household (measured in the migration module) and by maternal health at birth (vital statistics).
- A credible answer to a question the 2026 replication left open.

Assessment

Two problems, and they are decisive. **Already well covered.**

Age-at-exposure designs with birth location and cohort fixed effects: Bundervoet, Verwimp and Akresh (2009, *JHR*); Akresh, Bhalotra, Leone and Osili (2012, *AER P&P*); León (2012, *JHR*); Duque (2024, *JCR*); Akbulut-Yuksel, Tekin and Turan (2025, *JHR*). The prenatal-versus-postnatal angle is new; the design is not.

Too little variation. 70% of municipal end dates fall within eighteen months (July 2004 to December 2005). The identifying groups are thin: exposed only in utero means born in a nine-month window; never exposed means born after 2005, adults only from 2026 onwards. With current data, cohorts born after the end exist in 28 municipalities.

Verdict. A good question with a weak instrument. Not a job market paper unless a setting with more dispersed end dates, or linked birth records, is found.